Python Practice Set: Loops

Covers: variables, data types, print()/input(), string methods, if/elif/else, match-case, and for / while loops.

Part 1 — ATM PIN Attempt System (while loop)

Banks give you only a limited number of tries to enter your PIN correctly. Build that:

- Store a fixed correct PIN as a variable, e.g. correct_pin = "4521".
- Set a variable attempts = 0 and a maximum allowed attempts, e.g. max_attempts = 3.
- Use a while loop that keeps running as long as attempts is less than max_attempts.
- Inside the loop, ask the user to input a PIN.
- If it matches correct_pin, print "Login successful" and stop the loop (use break).
- If it doesn't match, increase attempts by 1, and print how many attempts remain.
- If the loop ends because attempts reached the maximum without success, print "Card blocked. Too many failed attempts."

Sample run to match:

```
Enter PIN: 1111
Wrong PIN. 2 attempt(s) remaining.
Enter PIN: 2222
Wrong PIN. 1 attempt(s) remaining.
Enter PIN: 4521
Login successful
```

> Hint: break immediately exits a loop, useful once the correct PIN is entered so the remaining attempts aren't wasted.

Part 2 — Daily Expense Tracker (for loop)

You want to log how much you spent each day for a week and see a summary at the end.

- Create a variable total_spent = 0 to keep a running sum.
- Use a for loop with range() that runs 7 times, once for each day (Day 1 to Day 7).
- On each loop pass, print which day it is (e.g. "Day 1"), then ask the user to input the amount spent that day, converting it to a float.
- Add that amount to total_spent on every loop pass.
- Inside the same loop, use an if/elif/else to label that day's spending: below 100 → "Low spending", 100 to 500 → "Moderate spending", above 500 → "High spending". Print the label right after the amount.
- After the loop ends, print the total spent for the week and the average (total_spent divided by 7).

Sample run to match:

```
Day 1 - Enter amount spent: 250
Moderate spending
Day 2 - Enter amount spent: 50
Low spending
...
Total spent this week: 1450.0
Average daily spend: 207.14
```

Hint: round(value, 2) keeps the average to two decimal places, which looks cleaner for money.

Part 3 — Password Strength Checker (for loop + strings)

Build a basic password strength checker using a for loop that goes character by character:

- Ask the user to input a password.
- Set up counters before the loop: has_digit = False, has_upper = False, has_lower = False.
- Loop through the password one character at a time using: for ch in password:
- Inside the loop, use string methods to check each character: ch.isdigit() to detect a digit, ch.isupper() to detect an uppercase letter, ch.islower() to detect a lowercase letter. Set the matching flag to True whenever found.
- After the loop finishes, check the password's length using len(password).
- Use if/elif/else to decide the strength: if length is less than 6 → "Weak" (regardless of anything else). Otherwise, if it has a digit AND an uppercase letter AND a lowercase letter → "Strong". Otherwise → "Medium".
- Print the final strength label.

Sample run to match:

```
Enter password: Hello123
Password strength: Strong

Enter password: hello
Password strength: Weak
```

Part 4 — Shopping Cart Loop with match-case (combined)

Combine everything: while loop, match-case, string methods, and running totals.

- Create total = 0 before the loop starts.
- Use a while True loop that keeps asking the user to input an item name, lowercased.
- If the user types "done", break out of the loop immediately and skip straight to printing the total.
- Otherwise, use match-case on the item name to set a price, same categories as before: "rice"|"wheat"|"dal" → 50, "milk"|"curd"|"paneer" → 60, "soap"|"shampoo" → 80, anything else (case _) → price 0 and print "Item not found, skipped."

- If the price isn't 0, add it to total and print a short confirmation like "Added rice - 50 rupees. Running total: 50".
- When the loop finally ends (user typed "done"), print the final total. If the total is above 300, also print "You unlocked a 10% discount!" and print the discounted total too.

Sample run to match:

```
Enter item (or 'done' to finish): rice
Added rice - 50 rupees. Running total: 50
Enter item (or 'done' to finish): milk
Added milk - 60 rupees. Running total: 110
Enter item (or 'done' to finish): soap
Added soap - 80 rupees. Running total: 190
Enter item (or 'done' to finish): done
Final total: 190
```

Checklist before you consider it done

- ATM system correctly blocks after the exact max number of wrong attempts, no more no less.
- Expense tracker loop runs exactly 7 times and total/average are correct.
- Password checker correctly flags a 5-character password as Weak even if it has digits and uppercase.
- Shopping cart loop exits cleanly when 'done' is typed, without trying to look up 'done' as an item.
- All loops use the correct type — while when the number of repeats is unknown upfront (PIN tries, shopping cart), for when it's fixed (7 days, characters in a password).